

HydroSOS Overview

Hydrological Forecasting, Data, Modelling, and Applications Training
Accra, Ghana, 27th – 29th January 2026



What is HydroSOS?

HydroSOS is the link between hydrological monitoring and decision making



What HydroSOS provides



An overview of the current hydrological status

including groundwater,
river flow and soil moisture



An appraisal of where the current status is significantly different from 'normal'

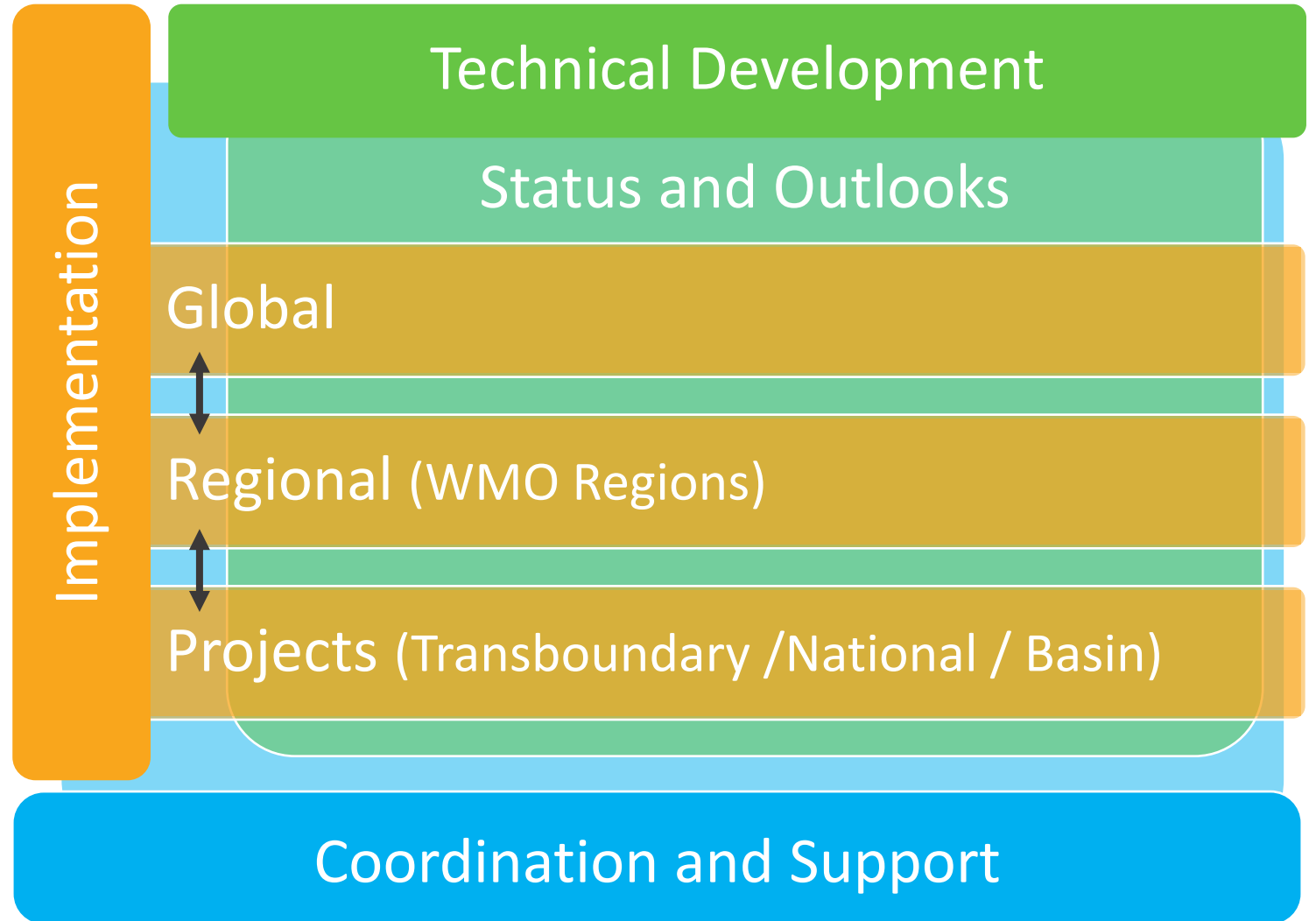
For example indicating
drought and flood susceptibility



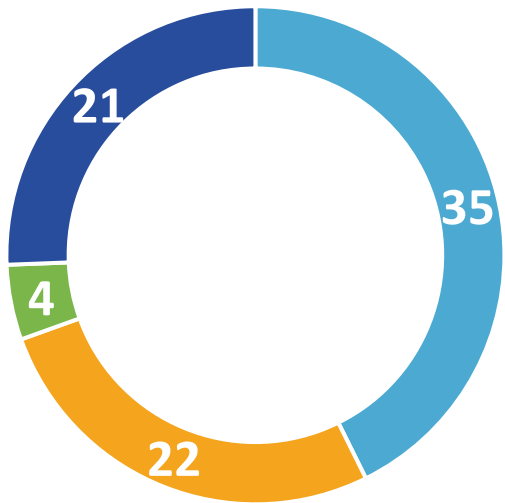
An assessment of whether this is likely to get better or worse

over coming weeks
and months

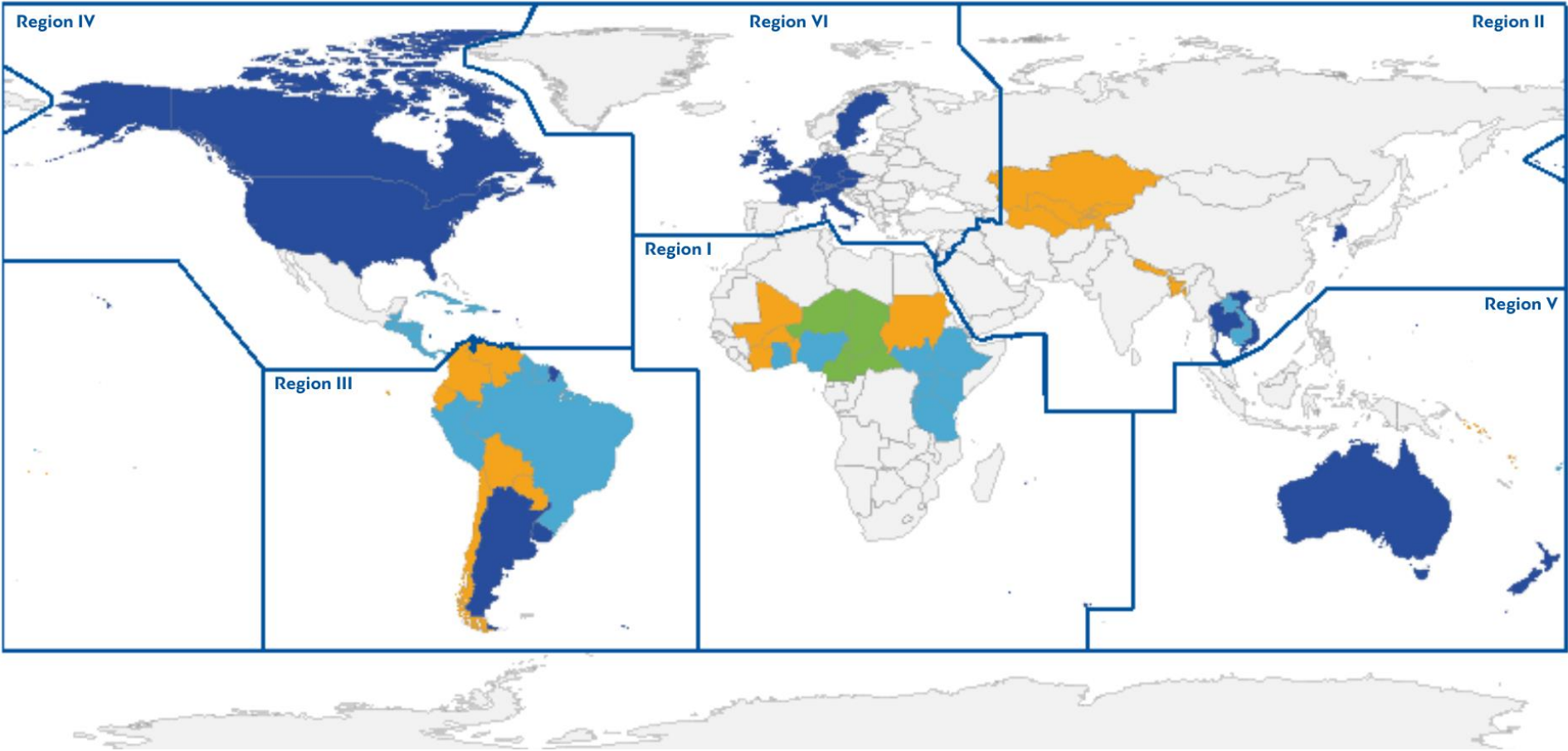
HydroSOS Structure



HydroSOS Implementation Status – 2025



Participating countries



HydroSOS Technical Developments

- **Interdisciplinary group of experts**
 - Lead: Lucy Barker, Katie Facer-Childs (UKCEH)
- **Methodologies and guidelines**
 - Streamflow status
 - Streamflow outlook
 - Groundwater status
 - Soil moisture status (under review)
 - Guidelines for Seasonal Hydrological Prediction
 - WIPPS Manual (S2S hydrological prediction section)
 - Definition of hydrological normal
 - **Upcoming: data infilling and QC, blending products**
 - **Upcoming: API / data transfer**
- **Tools and platforms**
 - [HydroSOS Portal](#)
 - [HydroSOS GitHub](#)
 - HydroSOS Plug-in for MCH (under review)
 - WHOS pilot phase and testing for data transfer
- **Capacity development**
 - 3-phased course:
 - Autonomous phase
 - Synchronous online phase (with trainer)
 - Face-to-face training(requires translation to English)
 - Modules on groundwater monitoring and rating curves available

Technical Developments

HydroSOS Implementation in Africa

- **WMO RA I - Africa**
 - RA I Hydrology and Water Coordination Panel. Chair: Mohamed Hussein
 - [Regional Implementation Plan](#)
- **Possible regional cooperation**
 - AGRHYMET
 - ACMAD
 - ECOWAS
 - African Development Bank (AfDB)
 - Adaptation Fund (AF)
 - World Bank (WB)
- **National Implementations**
 - Ghana
 - Nigeria
 - Togo
- **Regional Implementations**
 - Lake Chad
 - Lake Victoria & Nile Basin
 - Central Africa
 - Upcoming: AF Volta 2.0

Implementation

HydroSOS Implementation in Ghana

- **Identify groups, leads and roles**
- **Assess needs and current data, products, and services**
 - ie data infilling, remotely-sensed data, adaptation of hydro model for S2S, platforms, visualization, etc.
- **Prepare a workplan**
 - Identify human and financial resources
 - Gather all necessary data/inputs
 - Identify models and platforms
 - Define and develop products
 - Product dissemination
 - Etc.
- **Conduct capacity development**
- **Ensure maintenance and sustainability**
- **Example national products**
 - Historical conditions (i.e. past 24 months)
 - National and/or basin-specific hydrological Status Assessment (ie. first quarterly then monthly)
 - National and/or basin-specific Hydrological Outlooks (ie. first quarterly then monthly)
- **Contribution to regional products**
 - For example, through PRESASS

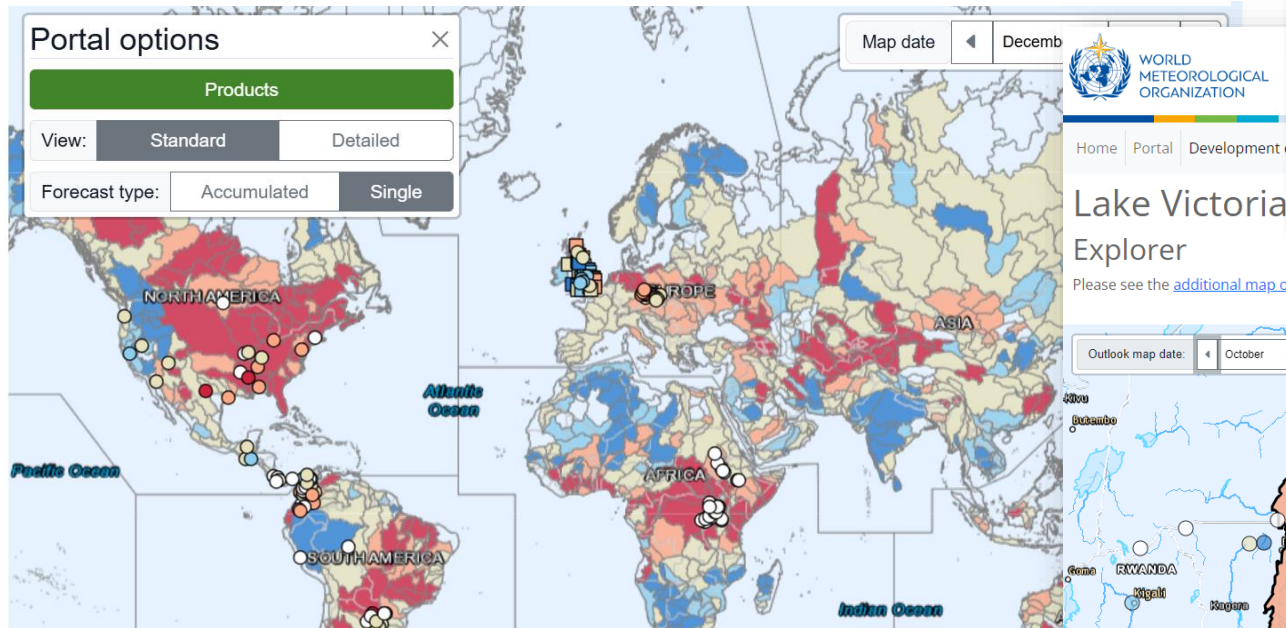
Implementation

HydroSOS capacity development in Africa

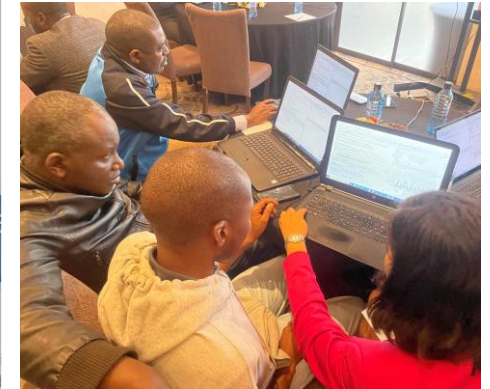


WMO Hydrological Status and Outlook System HydroSOS portal

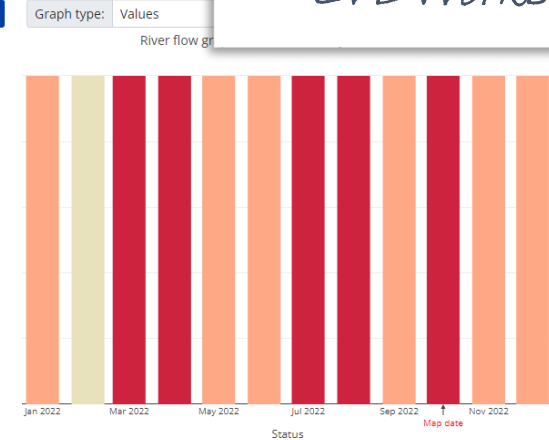
Home Portal About & help Case studies



HydroSOS
Ghana Workshop

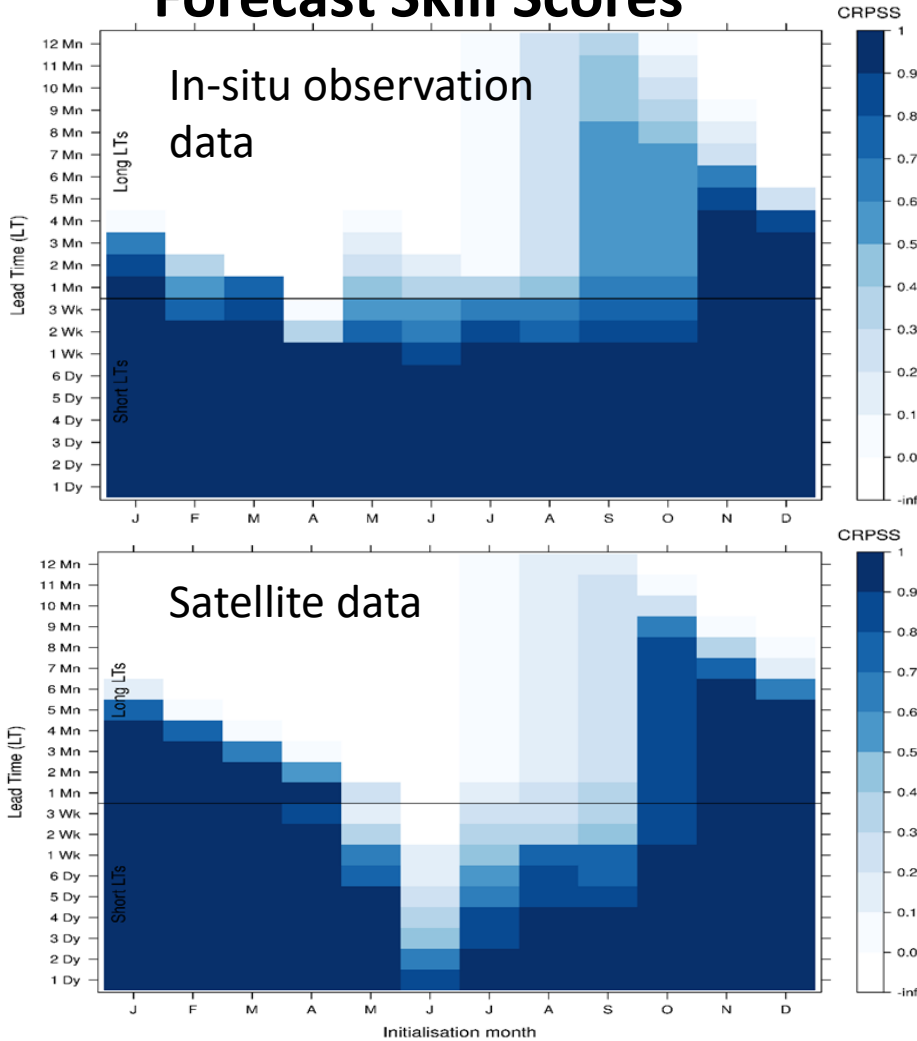


HydroSOS
LVB Workshop



Ghana - Sample Streamflow Outlook Results

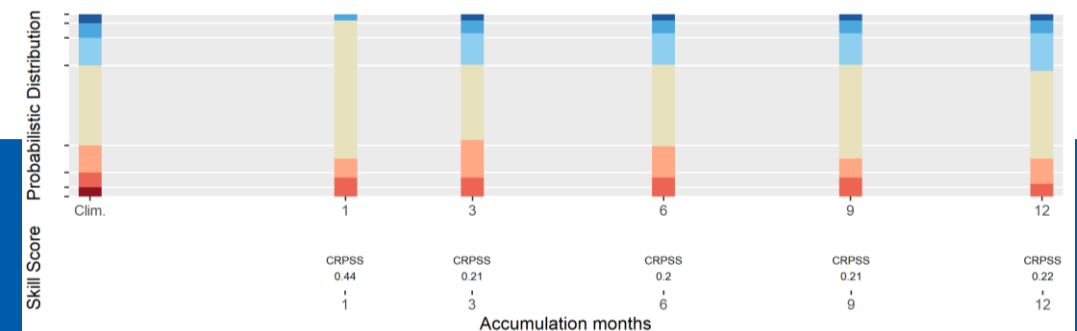
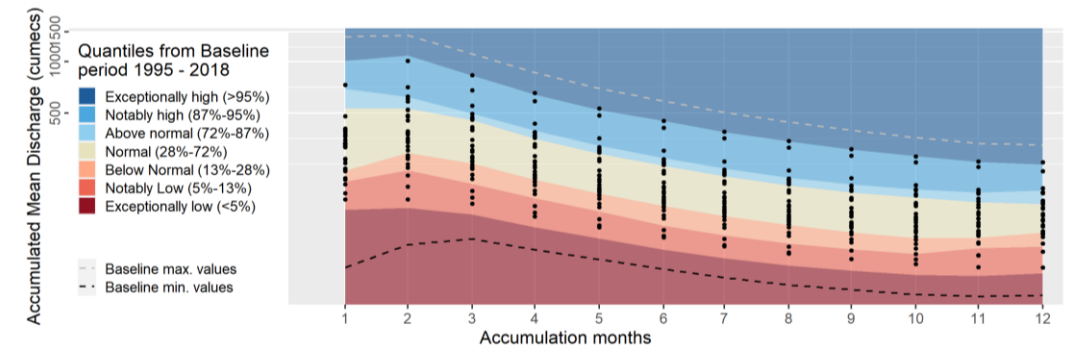
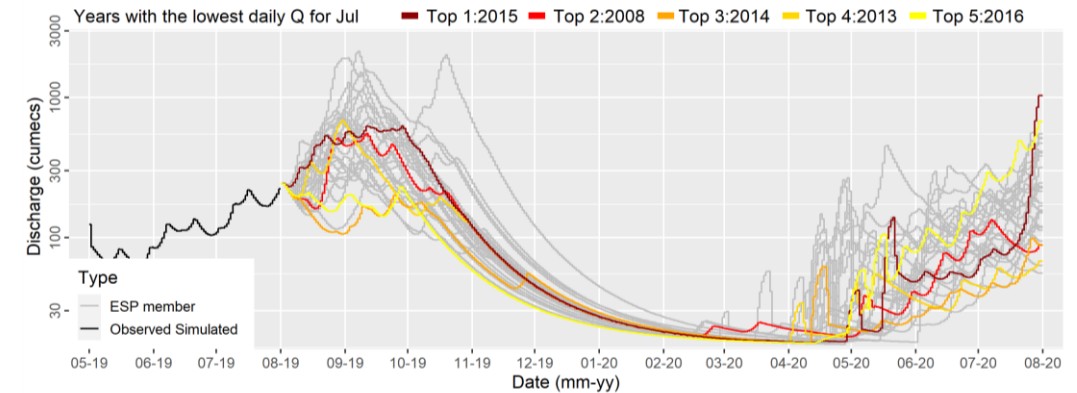
Forecast Skill Scores



Ensemble Streamflow Prediction Output

12-month ESP forecast from August 2019

Catchment: Pwalugu (1001001)



WORLD
METEOROLOGICAL
ORGANIZATION



HydroSOS example products



Example of National bulletin: Uruguay

Uruguay's Hydrological Bulletin

Hydrological
status for
national basins

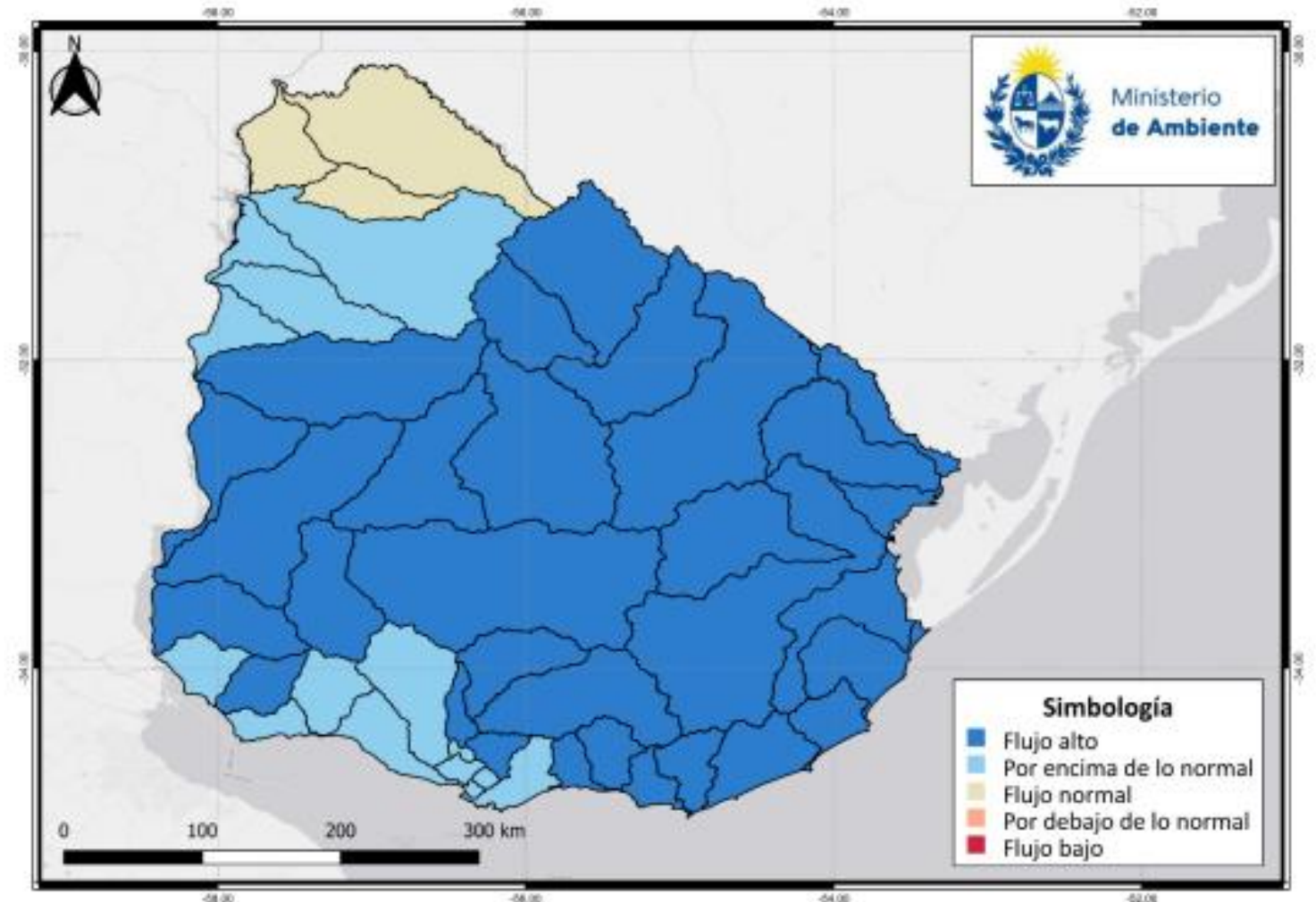


Figura 4. Mapa de categoría del estado hidrológico mensual para cuencas hidrográficas nivel 2.

Example of National bulletin: Uruguay

Uruguay's Hydrological Bulletin

Hydrological status
retrospective for
national basins

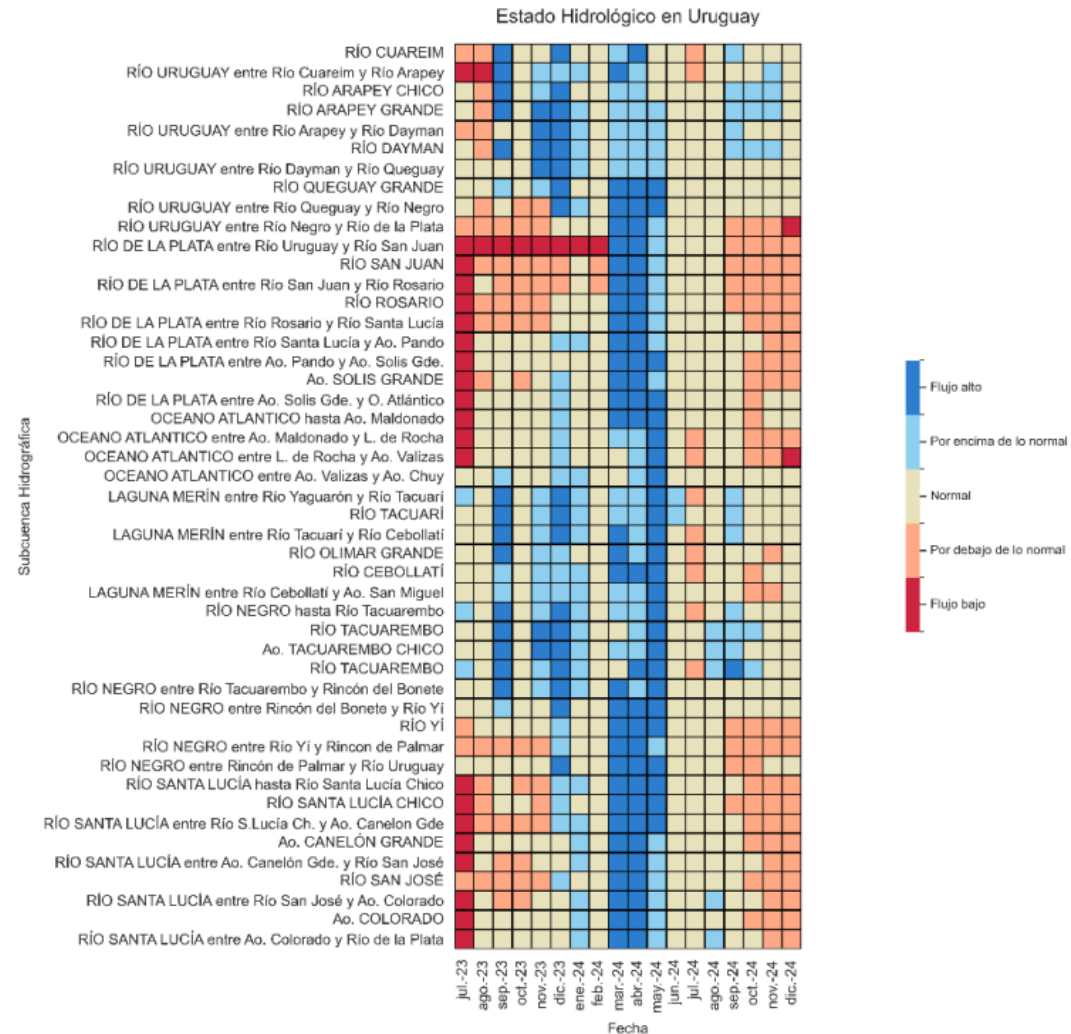


Figura 5. Estado hidrológico en cuencas hidrográficas para el periodo julio 2023 al diciembre 2024

Uruguay's Hydrological Bulletin

Hydrological outlook for sub-basins

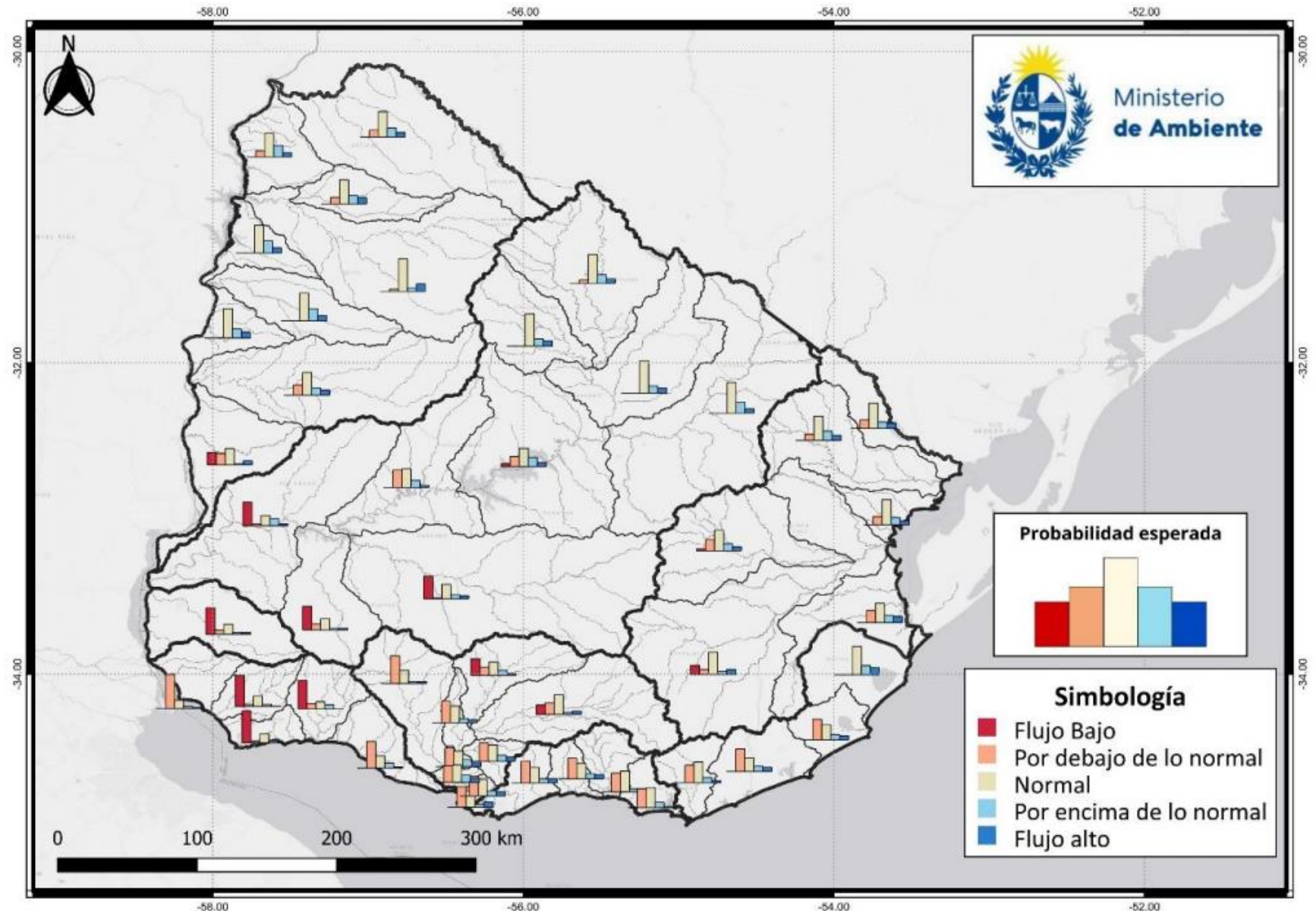


Figura 9. Perspectiva hidrológica para el mes de octubre 2024

Uruguay's Hydrological Bulletin

Hydrological outlook for sub-basins

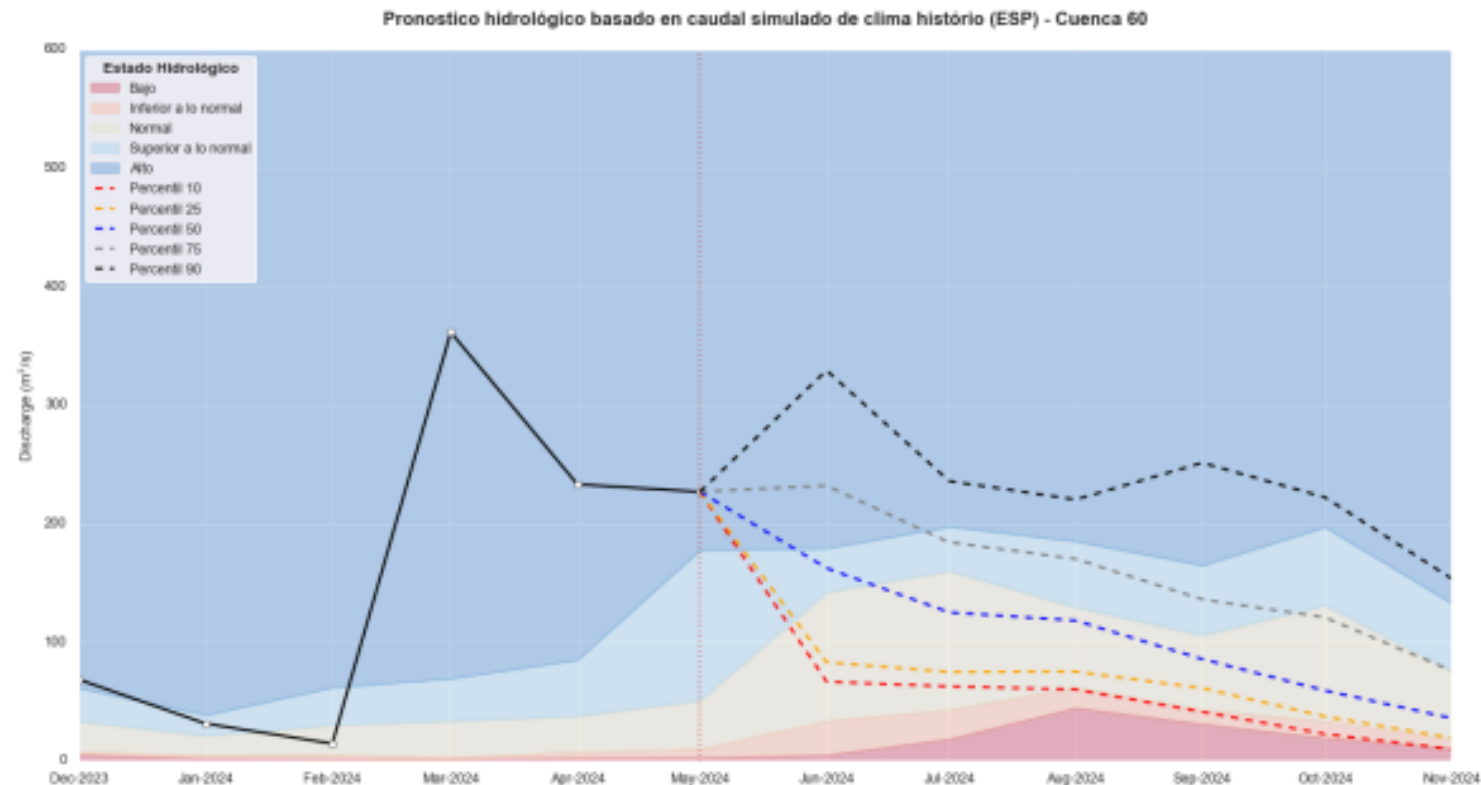


Figura 7. Perspectiva hidrológica para los próximos meses en la subcuenca del río Santa Lucía (código de cuenca 60).

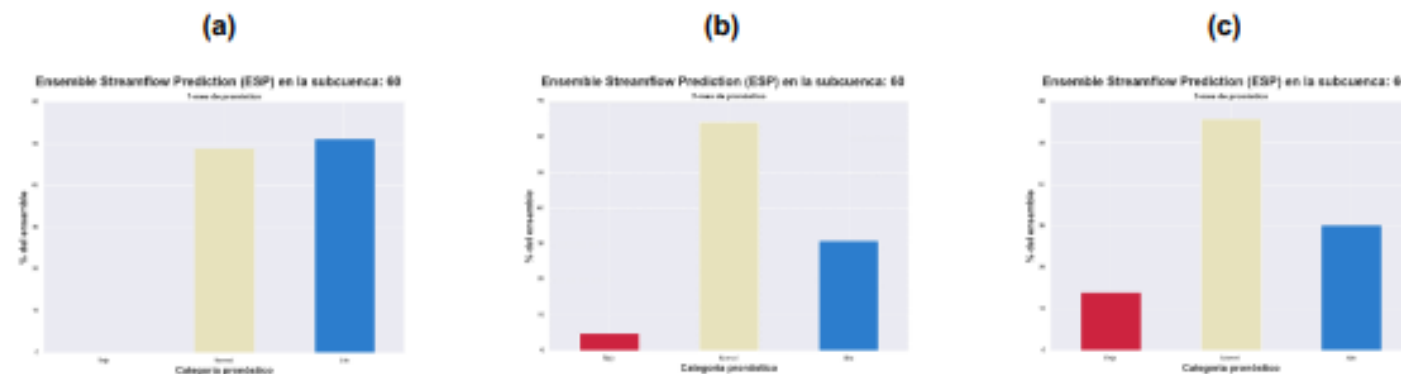
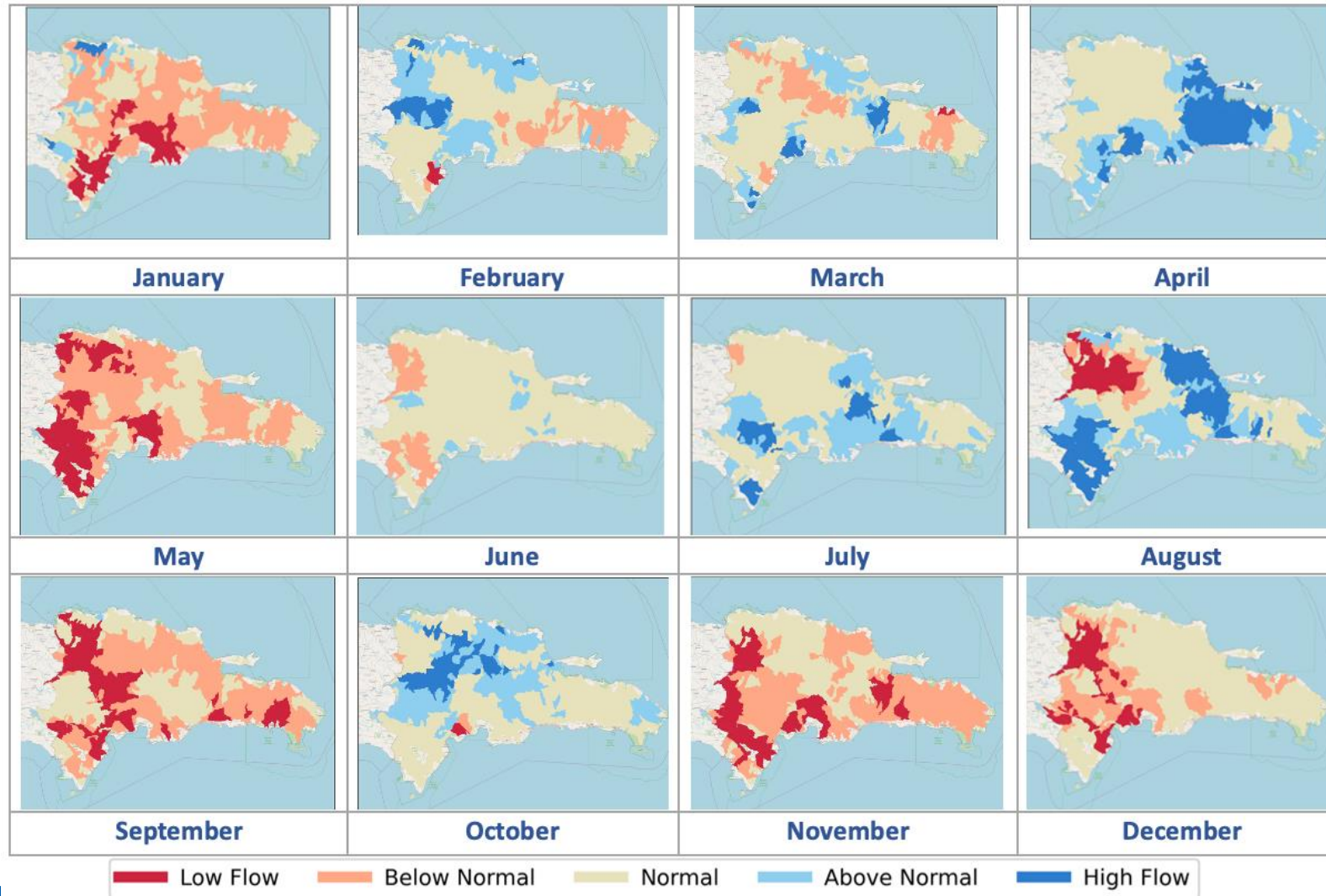


Figura 8. Perspectiva hidrológica de 1 mes (a), 2 meses (b), y 3 meses (c) en la subcuenca del río Santa Lucía (código cuenca 60).

Dominican Republic – Streamflow status



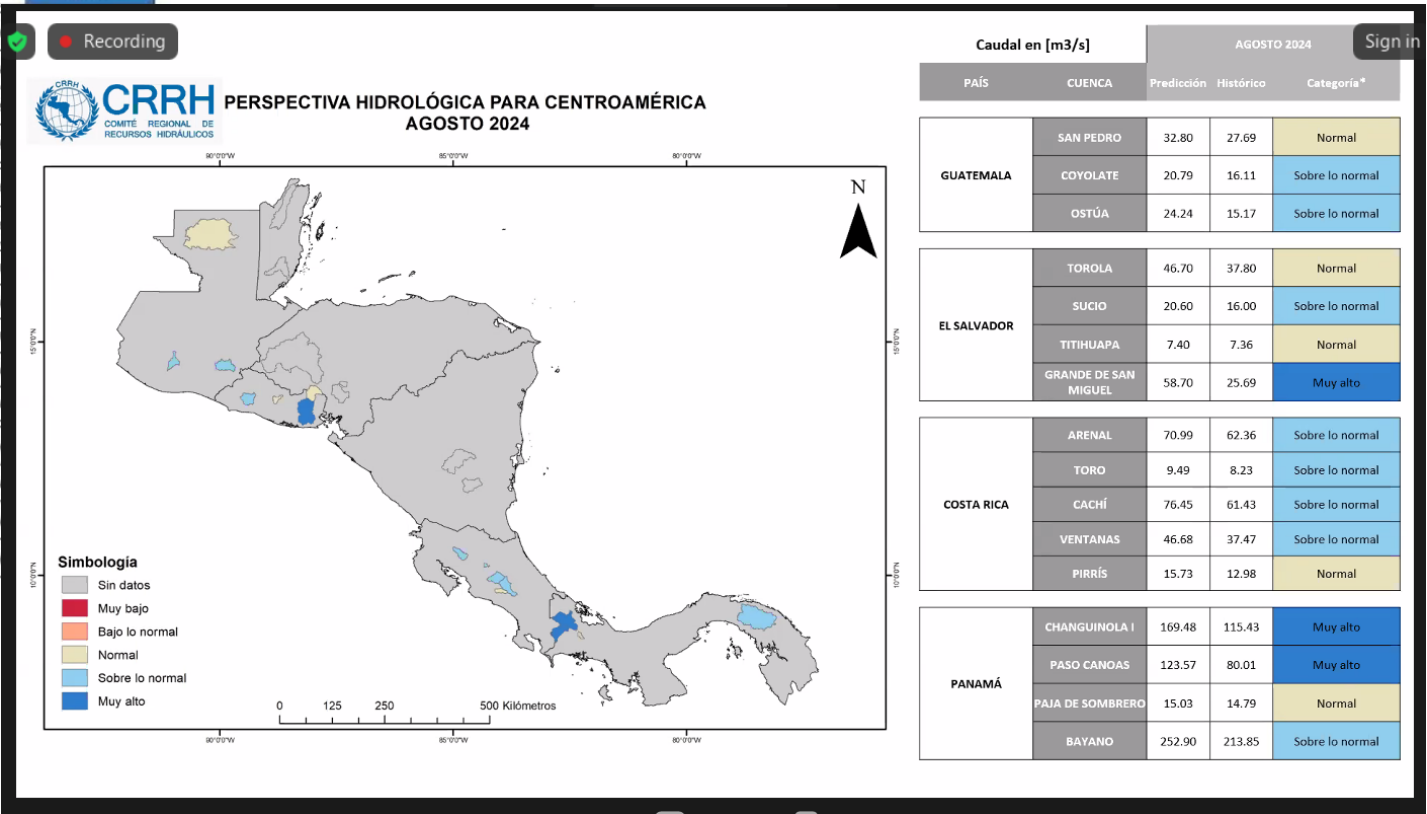
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HydroSOS as input to a regional bulletin

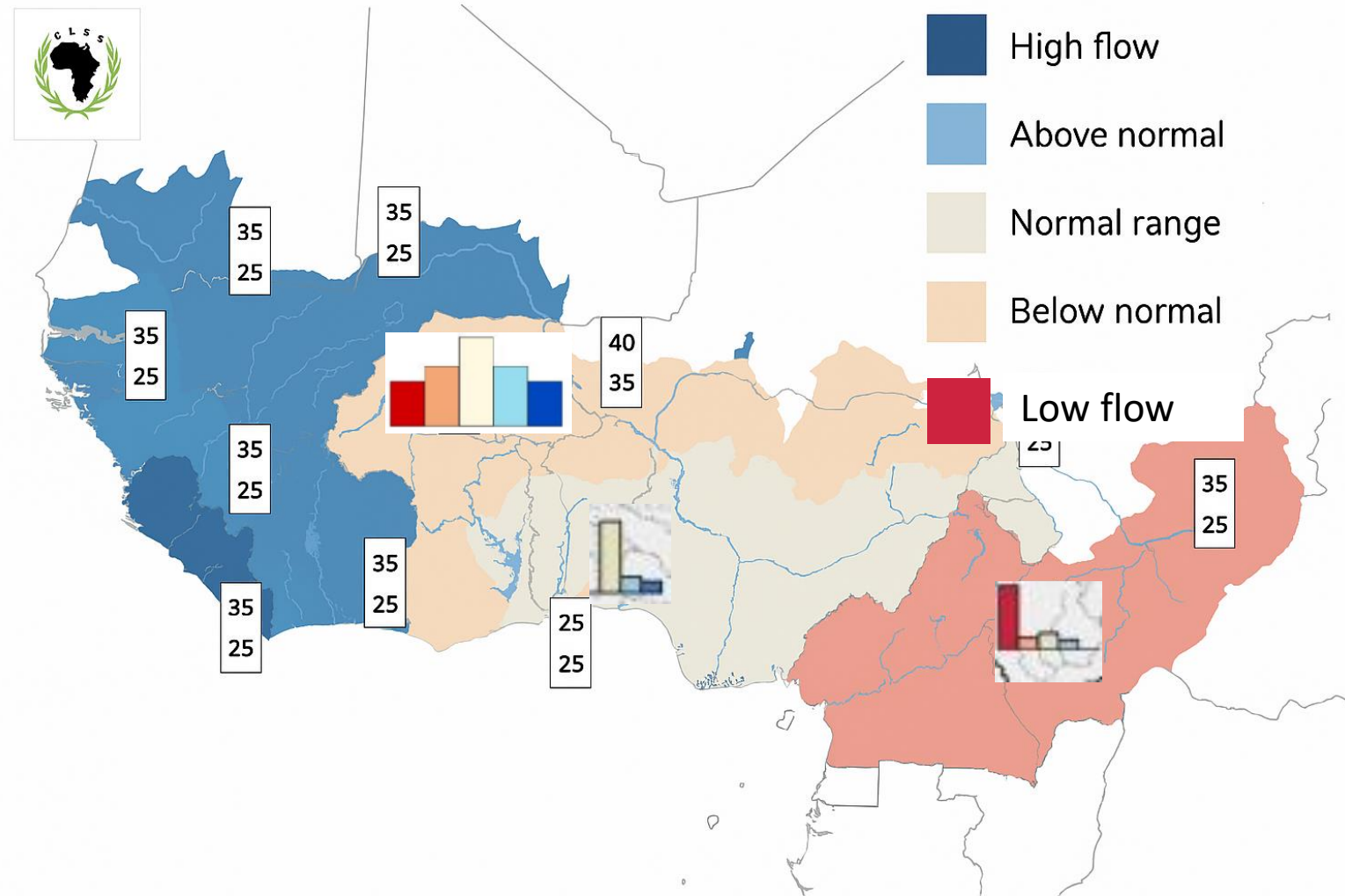
Caudal en m3/s		MAYO 2024				JUNIO 2024				JULIO 2024			
PAÍS	CUENCA	Predicción	Histórico	% Var.	Categoría*	Predicción	Histórico	% Var.	Categoría*	Predicción	Histórico	% Var.	Categoría*
GUATEMALA	SAN PEDRO	9.10	13.50	-77%	Muy bajo	8.80	26.80	-67%	Muy bajo	16.60	27.30	-39%	Normal
	COVOLATE	6.90	10.20	-36%	Muy bajo	15.20	15.70	-3%	Normal	21.70	14.20	53%	Muy alto
EL SALVADOR	TOROLA	25.10	18.10	39%	Sobre lo normal	39.10	56.80	-31%	Normal	40.10	30.60	31%	Muy alto
	SUCIO	5.70	7.60	-25%	Bajo lo normal	13.00	11.80	10%	Normal	19.90	12.30	62%	Muy alto
	TITIHUAPA	3.10	2.50	24%	Sobre lo normal	6.10	5.20	17%	Sobre lo normal	7.80	5.90	32%	Muy alto
	GRANDE DE SAN MIGUEL	31.10	17.90	74%	Normal	58.10	41.20	41%	Sobre lo normal	57.00	15.20	275%	Muy alto
NICARAGUA	SIGUA	12.13	16.40	-26%	Bajo lo normal	49.50	86.90	-43%	Muy bajo	134.50	180.20	-25%	Muy alto
	VALENTIN	8.05	13.22	-39%	Muy bajo	40.25	56.06	-29%	Muy bajo	83.80	99.62	-16%	Muy alto
COSTA RICA	ARENAL	27.11	30.57	-11%	Normal	39.05	45.33	-14%	Normal	42.57	60.57	-30%	Muy alto
	TORO	5.13	5.55	-8%	Normal	9.47	7.08	34%	Muy alto	9.41	8.30	13%	Muy alto
	CACHÍ	29.77	41.80	-29%	Bajo lo normal	49.68	58.13	-15%	Normal	47.65	58.19	-18%	Muy alto
	VENTANAS	22.15	22.45	-1%	Normal	47.36	35.17	35%	Sobre lo normal	42.64	31.51	35%	Muy alto
	PIRRÍS	7.94	6.17	22%	Sobre lo normal	13.28	11.41	16%	Sobre lo normal	13.25	10.22	30%	Muy alto
PANAMÁ	CHANGUINOLA I	158.02	132.94	19%	Sobre lo normal	215.10	159.90	37%	Sobre lo normal	198.96	152.18	30%	Muy alto
	PASO CANOAS	59.75	56.53	6%	Normal	85.31	78.05	22%	Sobre lo normal	100.70	71.87	42%	Muy alto
	PAJA DE SOMBRERO	15.35	9.46	62%	Muy alto	28.14	14.49	94%	Muy alto	16.67	13.36	25%	Muy alto
	BAYANO	136.90	164.78	-17%	Bajo lo normal	183.90	191.84	-4%	Normal	188.00	194.72	-4%	Muy alto

Central American Hydrological Outlook



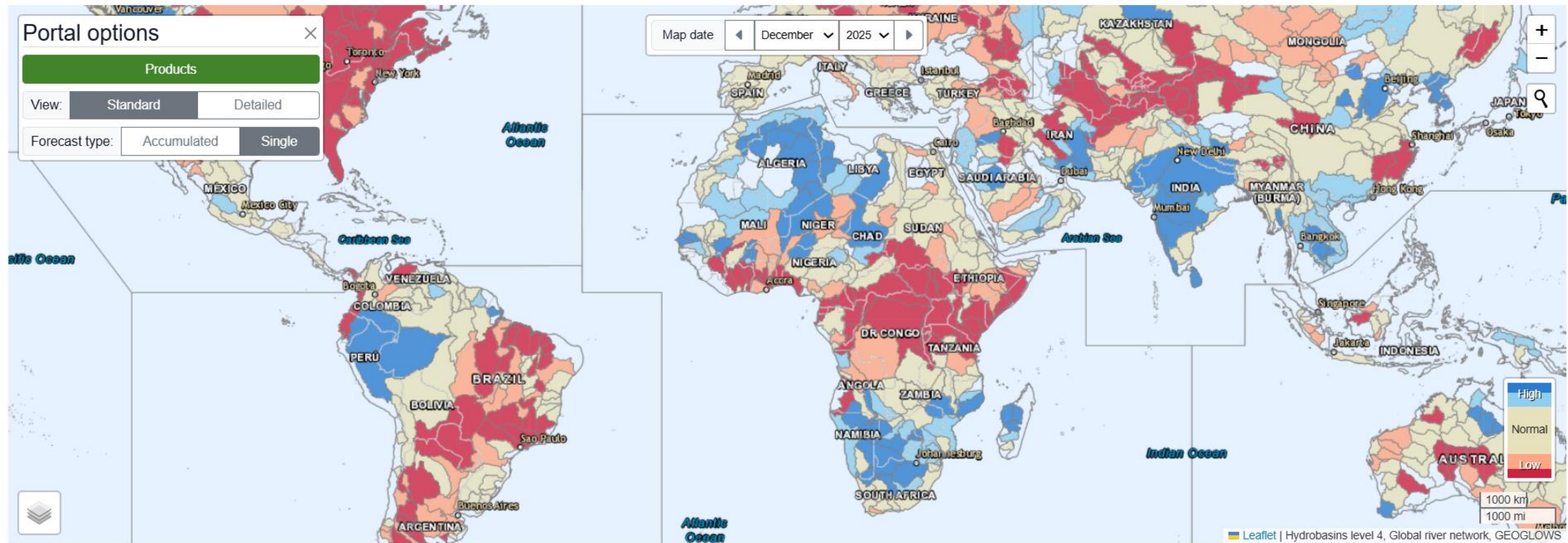
For West Africa, that could translate into something like this:

Mock-up



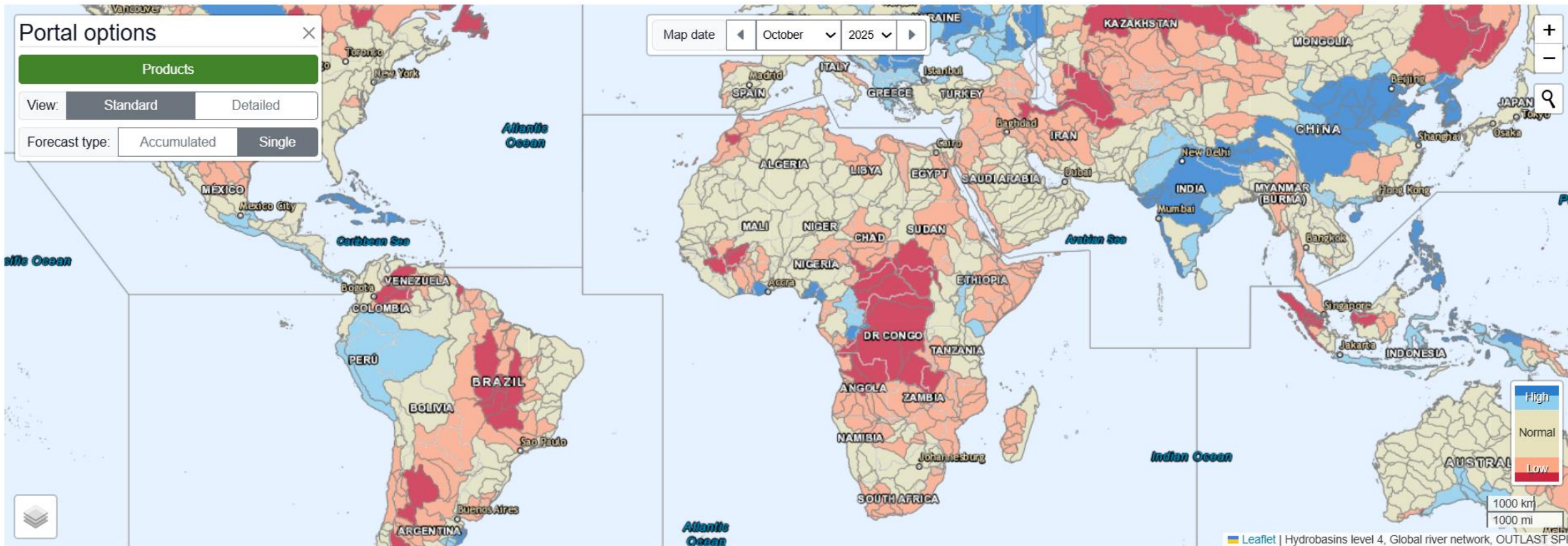
HydroSOS Portal – GEOGLOWS Global Product

<https://wmohydrosos.ceh.ac.uk/portal>



HydroSOS Portal – Global SPI

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Thank you

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